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RV COLLEGE OF ENGINEERING®

Autonomous Institution affiliated to VTU

VI Semester, B.E.

Model Question Paper (2022 SCHEME)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (Data Science)

COURSE TITLE: BIG DATA SYSTEMS

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10
- 3.

PART-A

1	1.1	_____ data is stored in tables with rows and columns, typically in relational databases.	1
	1.2	JSON and XML are examples of _____ data.	1
	1.3	True/False: Unstructured data can be easily stored and queried in a relational database.	1
	1.4	True/False: A B-tree index improves disk locality by keeping related data close together.	1
	1.5	How does in-memory processing reduce latency compared to disk-based processing?	1
	1.6	_____ processing divides a task across multiple CPUs/cores to speed up computation.	1
	1.7	The _____ model requires processes to communicate by sending and receiving messages.	1
	1.8	True/False: Distributed systems always have lower latency than single-machine systems.	1
	1.9	Give an example of a system optimized for Availability (AP) under CAP.	1
	1.10	Hadoop is designed to store and process _____ datasets across distributed clusters.	1
	1.11	The _____ is the master node in HDFS that manages metadata.	1
	1.12	True/False: Hadoop 1.x uses YARN for resource management.	1
	1.13	Explain the role of ZooKeeper in Hadoop	1
	1.14	What is Oozie's role in Hadoop?	1
	1.15	True/False: Flume is used for real-time data streaming like Kafka.	1
	1.16	HPCC stands for _____.	1
	1.17	_____ is the declarative programming language used in HPCC	1
	1.18	True/False: Thor is optimized for batch processing, while ROXIE handles real-time queries.	1
	1.19	Spark is designed for _____ processing, unlike Hadoop's batch-only approach.	1
	1.20	The _____ in Spark allows for fault tolerance by tracking lineage of transformations	1

PART-B

2	a	Discuss different types of data and their associated storage systems. Explain structured, semi-structured, and unstructured data with examples.	08
	b	Describe the principle of locality of reference with examples. How does it affect data access performance?	08

3	a	Explain the motivation for parallel and distributed processing in Big Data. Compare shared memory with message passing architectures.	08
	b	Discuss data access strategies in distributed systems. Explain partitioning, replication, and messaging with examples.	08
		OR	
4	a	Compare Client-Server and Peer-to-Peer models. Explain cluster computing architecture and its components.	08
	b	Explain the CAP Theorem and its relevance to Big Data analytics. Discuss consistency, availability, and partition tolerance with examples.	08

5	a	Explain Hadoop's architecture. Describe how MapReduce and HDFS work together in data processing.	08
	b	Discuss the role of Hadoop ecosystem tools like HBase, Hive, Pig, Sqoop, Flume, Zookeeper, and Oozie.	08
OR			
6	a	Give an example of a Hadoop application and explain its workflow.	08
	b	Write a MapReduce program in Java to count word frequencies in a text file.	08

7	a	Explain HPCC Systems and its architecture. Compare Thor and ROXIE. How does it differ from Hadoop?	08
	b	Describe ECL programming and its components. Provide examples for records, functions, and modules.	08

OR

8	a	Explain data shaping operations in HPCC like Sort, Join, Normalize, Denormalize, and GROUP functions.	08
	b	Write an ECL script to join two datasets (Customers and Orders) and compute total sales per customer.	08

9	a	Describe Apache Spark's architecture and features. How does Spark handle in-memory computation?	08
	b	Explain Spark transformations with examples. How are RDDs used in data processing?	08

OR

10	a	Discuss machine learning and streaming with Spark. Provide examples of regression, clustering, and stream processing.	08
	b	Explain the micro-batch architecture of Spark Streaming with a diagram.	08

Signature of Scrutinizer:

Signature of Chairman

Name:

Name: